

# Example: From Transactional DB to Ordered Frequent Itemlist

Example: A Sample Transactional Database

| TID | Items in the Transaction     |
|-----|------------------------------|
| 100 | $\{f, a, c, d, g, i, m, p\}$ |
| 200 | $\{a, b, c, f, l, m, o\}$    |
| 300 | $\{b, f, h, j, o, w\}$       |
| 400 | $\{b, c, k, s, p\}$          |
| 500 | $\{a, f, c, e, l, p, m, n\}$ |

Let min\_support = 3

- Scan DB once, find single item frequent pattern:  $f:4, a:3, c:4, b:3, m:3, p:3$
- Sort frequent items in frequency descending order, f-list  $F\text{-list} = f\text{-}c\text{-}a\text{-}b\text{-}m\text{-}p$
- Scan DB again, use the ordered frequent itemlist for each transaction to construct an FP-tree

| TID | Items in the Transaction     | Ordered, frequent itemlist |
|-----|------------------------------|----------------------------|
| 100 | $\{f, a, c, d, g, i, m, p\}$ | $f, c, a, m, p$            |
| 200 | $\{a, b, c, f, l, m, o\}$    | $f, c, a, b, m$            |
| 300 | $\{b, f, h, j, o, w\}$       | $f, b$                     |
| 400 | $\{b, c, k, s, p\}$          | $c, b, p$                  |
| 500 | $\{a, f, c, e, l, p, m, n\}$ | $f, c, a, m, p$            |

# Example: Construct FP-tree from Transaction DB

| TID | Ordered, frequent itemlist |
|-----|----------------------------|
| 100 | <i>f, c, a, m, p</i>       |
| 200 | <i>f, c, a, b, m</i>       |
| 300 | <i>f, b</i>                |
| 400 | <i>c, b, p</i>             |
| 500 | <i>f, c, a, m, p</i>       |

## FP-Tree Construction:

For each transaction, insert the ordered frequent itemlist into an FP-tree, with shared sub-branches merged, counts accumulated

After inserting the 1<sup>st</sup> frequent Itemlist: "f, c, a, m, p"

| Item | Frqncy | hdr |
|------|--------|-----|
| f    | 4      |     |
| c    | 4      |     |
| a    | 3      |     |
| b    | 3      |     |
| m    | 3      |     |
| p    | 3      |     |

Header Table

After inserting the 2<sup>nd</sup> frequent itemlist "f, c, a, b, m"

| Itm | hdr |
|-----|-----|
| f   |     |
| c   |     |
| a   |     |
| b   |     |
| m   |     |
| p   |     |

After inserting all the frequent itemlists

| Itm | hdr |
|-----|-----|
| f   |     |
| c   |     |
| a   |     |
| b   |     |
| m   |     |
| p   |     |